

Ahmed Arfaoui

AI & Data Science Engineer

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Profile

AI engineer specializing in agentic systems, browser automation, and end-to-end machine learning products. Built and evaluated multi-agent workflows with LangGraph, LangChain, MCP, n8n, FastAPI, Next.js, and PostgreSQL, combining production software engineering with NLP, computer vision, MLOps, and measurable experimentation.

Professional Experience

AI & Agentic Systems Engineering Intern | Soft Stars

Dec 2025 - Jun 2026

PFE Project: Open Web Catcher (OWC)

- Designed and built OWC, a reviewable multi-agent platform that collects browser evidence from dynamic streaming sites.
- Created LangGraph orchestration and specialist agents for classification, landing discovery, player activation, iframe extraction, provider analysis, and human-reviewed email drafting; bound scoped tools through LangChain and MCP.
- Built an n8n/Puppeteer prototype that reached 150+ working websites and deployed the landing agent as an operational backup for dynamic targets.
- Validated 126 persisted runs: 73.7% strict success, 97.6% tool-call success, 217 stream records across 24 evidence-producing sites, and \$0.152 average model cost per run.

Data Science Intern | VERMEG

Aug 2025 - Sep 2025

Legacy Code Modernization with Spring AI

- Automated migration of 50+ legacy XML security configurations to Java/Spring with Spring AI, reaching 90% conversion accuracy.
- Supported default and custom schemas, reducing manual refactoring by 95% and shortening modernization from weeks to days.

Data Science Intern | ESPRIT

Jun 2025 - Aug 2025

AI-Driven Student Group Formation

- Engineered LLM-generated assessments and benchmarked six clustering algorithms; K-Means achieved a 15% higher silhouette score.
- Served the Python clustering pipeline through FastAPI with an average response time under 200 ms.

BI & Data Science Intern | CMR Tunisie

Jul 2024 - Aug 2024

Sales Analytics and Forecasting

- Built two Power BI dashboards and ARIMA, SARIMA, SARIMAX, and Prophet forecasting models for sales decision support.

Selected Projects

NEWSBOT AI

Sep 2025 - Nov 2025

Built a RAG pipeline with FAISS and Sentence Transformers, fine-tuned FLAN-T5 with LoRA, and added trend, sentiment, bias, SHAP, and LIME analysis.

FarmWise

Jan 2025 - May 2025

Reached 95% ResNet plant-disease accuracy and 88% YOLOv8 mAP; added RAG over 1,000+ documents in a Next.js/Django platform.

End-to-End MLOps Pipeline

Sep 2024 - Jan 2025

Combined MLflow, DVC, Docker/FastAPI, CI/CD, Elasticsearch, and Kibana; reduced commit-to-artifact time by over 80%.

Speech Emotion Recognition

Sep 2024 - Jan 2025

Used diffusion-based mel-spectrogram augmentation to raise EmoDB emotion-recognition accuracy from 82.1% to 94.3%.

Telecom Customer Churn Prediction

Sep 2024 - Dec 2024

Compared GBM and Random Forest models, applied SMOTE for class imbalance, and used SHAP to explain churn drivers; the best pipeline reached 92% accuracy.

Technical Skills

Agentic AI: LLM agents, agent creation, ReAct, prompt engineering, n8n, LangChain, LangGraph, MCP, RAG, LoRA, FAISS

AI & Machine Learning: Python, PyTorch, TensorFlow/Keras, scikit-learn, NLP, computer vision, YOLOv8, XAI, SHAP, LIME

Engineering & Data: FastAPI, Django, Spring Boot, Next.js, PostgreSQL, SQL, Power BI, Java, JavaScript

MLOps & Automation: Docker, MLflow, DVC, CI/CD, AWS, Git, Linux, Elasticsearch, Kibana, Puppeteer, Playwright

Education

ESPRIT: Engineering Degree in Software Engineering, specialization in Data Science & AI (2026)

IPEIB: Preparatory Cycle for Engineering Schools - Mathematics & Physics

Tinja High School: Baccalaureate in Mathematics

Certifications & Languages

Certifications: AWS Academy Cloud Foundations; Hashgraph Developer Course; NVIDIA Transformer-Based NLP Applications

Languages: Arabic (native); French (fluent); English (fluent)